

torch.cross#

<https://pytorch.org/docs/stable/generated/torch.cross.html>

Description:

Returns the cross product of vectors in dimension dim of input and other. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of vectors, for which it computes the product along the dimension dim. In this case, the output has the same batch dimensions as the inputs. If dim is not given, it defaults to the first dimension found with the size 3. Note that this might be unexpected. This behavior is deprecated and will be changed to match that of `torch.linalg.cross()` in a future release.

Function Signature

```
torch.cross(input, other, dim=None, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(4, 3)
>>> a
tensor([[ -0.3956,  1.1455,  1.6895],
        [ -0.5849,  1.3672,  0.3599],
        [ -1.1626,  0.7180, -0.0521],
        [ -0.1339,  0.9902, -2.0225]])
>>> b = torch.randn(4, 3)
>>> b
tensor([[ -0.0257, -1.4725, -1.2251],
        [ -1.1479, -0.7005, -1.9757],
        [ -1.3904,  0.3726, -1.1836],
        [ -0.9688, -0.7153,  0.2159]])
>>> torch.cross(a, b, dim=1)
tensor([[ 1.0844, -0.5281,  0.6120],
        [-2.4490, -1.5687,  1.9792],
        [-0.8304, -1.3037,  0.5650],
        [-1.2329,  1.9883,  1.0551]])
>>> torch.cross(a, b)
tensor([[ 1.0844, -0.5281,  0.6120],
        [-2.4490, -1.5687,  1.9792],
        [-0.8304, -1.3037,  0.5650],
        [-1.2329,  1.9883,  1.0551]])
```

Notes & Warnings

WARNING

Warning If `dim` is not given, it defaults to the first dimension found with the size 3. Note that this might be unexpected. This behavior is deprecated and will be changed to match that of `torch.linalg.cross()` in a future release.

NOTE

See also `torch.linalg.cross()` which has `dim=-1` as default.

Detailed Documentation

Documentation

Returns the cross product of vectors in dimension `dim` of `input` and `other`.

Supports input of `float`, `double`, `cfloat` and `cdouble` dtypes. Also supports batches of vectors, for which it computes the product along the dimension `dim`. In this case, the output has the same batch dimensions as the inputs.

If `dim` is not given, it defaults to the first dimension found with the size 3. Note that this might be unexpected.

This behavior is deprecated and will be changed to match that of `torch.linalg.cross()` in a future release.

`torch.linalg.cross()` which has `dim=-1` as default.

`input` (Tensor) – the input tensor.

`other` (Tensor) – the second input tensor

`dim` (int, optional) – the dimension to take the cross-product in.

`out` (Tensor, optional) – the output tensor.